

Everyone is Trying to Automate Research

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Introduction

The paradigm of scientific discovery is undergoing a structural shift. Driven by breakthroughs in Large Language Models (LLMs), agentic orchestration, tool integration (e.g., Model Context Protocol), and automated code/sandbox execution, AI use in research is transitioning rapidly from a passive literature assistant to an autonomous co-scientist.

The use of generative AI in the automation of research began in 2025 and continues relentlessly. This begs the important question asked by Arvind Narayanan in a recent presentation – “[What will be left for us to work on?](#)”. I urge you to read through his presentation, it offers a useful framework for thinking about AI automation of the world, not just research. It’s based on the philosophy that AI is yet another normal technology ([Narayanan and Kapoor 2025](#)) that will diffuse through the economy albeit in a very different disruption pattern than others. As we may imagine, automating research will call for systems that aid humans in understanding the generated results. This process of understanding will also include a process of verification and evaluation of the work done by AI agents.

There is already an extensive debate around the role of humans in research done with AI. It seems to devolve around the fact that human understanding is sacrosanct. Even for results produced by AI, we need to make sure that the human researcher deeply understands how those results were achieved and is able to place them within the pantheon of existing results in a specific domain. Some of this will come from the use of automated tools like Lean4 in the mathematical realm, which takes care of the *verification* step, but has not matured into handling the *understanding* step as yet.

In this article, we do not discuss these issues around human understanding. Here, we aim to understand the variety of work that has been done to automate research. After that, we’ll also look at whether this means the end of academic research as we know it.

The Evolution of Automation in Research

AI-assisted Research automation has moved through five stages in less than that many years.

How will AI change science?

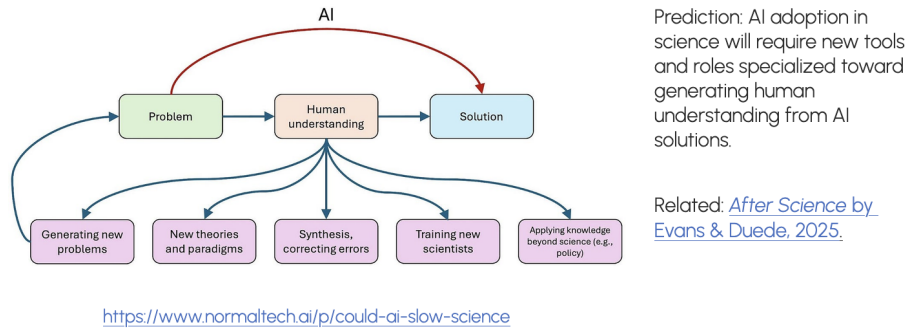


Figure 1: AI solutions will need human understanding (from the talk by Arvind Narayanan, linked above.)

1. Task-specific Assistance. Here we see AI help us in code generation to undertake empirical analyses (Aygün et al. 2025), or in surveying papers for related literature, solve open problems, refute conjectures, and generate new proofs across diverse areas in theoretical computer science, as well as other areas such as math, economics, optimization, and physics (Woodruff et al. 2026).
2. End-to-End Open-Ended Discovery. Instead of specific tasks, end to end research work involves literature review, hypothesis generation, code implementation, execution, and manuscript composition. This is one of the most popular and common modes of automated research. (Mitchener et al. (2025), Weng et al. (2025), Lu et al. (2024), Yamada et al. (2025), Tang et al. (2025)).
3. Research with Grounding and Verifiability. These are frameworks addressing hallucination, divergence between code and the paper, and handling of statistical replication errors. Chain of Evidence (CoE) traces every claim to a verifiable source (Meng et al. 2026). Curie (Kon et al. 2025) embeds rigor into the experimentation process through three components: an intra-agent rigor module to enhance reliability, an inter-agent rigor module to maintain methodical control, and an experiment knowledge module to enhance interpretability. Research problem formulation uses approaches like Structural Gap Hypothesis Agent (SGHA), (Gharat and Komiyama 2026)
4. Creating Interactive Research Agents. An approach that shifts research papers from static PDFs to interactive agents that enable human-AI

research workflows. Miao et al. (2025) converts papers automatically into research agents. Using tools such as AgentRxiv (Schmidgall and Moor 2025), research agents are able to reason better and generate higher quality research over a baseline. Current AI systems struggle to fully engage with human research because they fail to model essential scientific infrastructure like peer review, collaboration, and structured knowledge networks. To resolve this issue, (Shao et al. 2025) embeds these real-world scientific mechanisms directly into the AI workflow. This integration enables a more genuine research ecosystem that can interact deeply with the scientific community.

5. Self-Evolving Multi-Agent Systems for Research. This research approach builds pipelines that accumulate long-term memory, learn from failed experiments, and iteratively improve search strategies. EvoScientist (Lyu et al. 2026) is a multi-agent AI framework that continuously improves scientific research through persistent memory modules for ideation and experimentation, managed by three specialized agents (Researcher, Engineer, and Evolution Manager). By retrieving past successes and avoiding prior failures, the system progressively enhances both idea generation and code execution. Consequently, EvoScientist outperforms leading open-source and commercial AI systems across novelty, feasibility, relevance, and clarity. AutoResearchClaw (Liu et al. 2026) is a multi-agent autonomous framework that addresses the limitations of linear AI pipelines by incorporating multi-agent debate, self-healing execution loops, verifiable reporting, human collaboration, and cross-run memory. Through these mechanisms, it outperforms AI Scientist v2 by 54.7% on ARC-Bench while demonstrating that targeted human intervention at key decision points produces the best research outcomes. PiFlow (Pu et al. 2025) is an information-theoretical framework that treats automated scientific discovery as a structured, principle-guided uncertainty reduction problem, acting as a plug-and-play module for existing agent architectures. Across three domains, it accelerates time-to-solution by 5.6x, reduces token consumption by up to 27%, and boosts both discovery efficiency and solution quality over state-of-the-art methods.

The earlier automated research agents used a sequential architecture, comprising a paradigm of *Idea -> Code -> Run -> Paper*, see for example QuickAgent-Researcher. This has evolved into complex, search-based, and evolutionary architectures with some interesting features and abstractions. (i) *Agentic tree search* emerged as one architecture, where systems like AI Scientist-v2 (Yamada et al. 2025) replace linear workflows with Best-First Tree Search (BFTS) or Monte Carlo Tree Search (MCTS) over hypothesis spaces. This allows the system to backtrack when experiments yield null results or code execution fails, branching out into alternate hypothesis trees. (ii) Sequential architectures are also not good at verification and looping back; this is addressed in approaches that use *Chain of Evidence* (CoE, Meng et al. (2026)), where “method-code divergence” (describing an algorithm in text that differs from the underlying Python code) or

hallucinated bibliographies is mitigated. ScientistOne ensures that every claim in a generated paper must mathematically and procedurally trace back to verified execution logs, raw JSON metrics, and validated citation IDs. (iii) *Managing memory* is another important aspect of automated research systems. EvoScientist (Lyu et al. 2026) introduces dual-memory modules: Ideation Memory (recording feasible vs. infeasible hypothesis paths) and Experimentation Memory (capturing environment setup fixes, hyperparameter dynamics, and debugging tactics). The system evolves across research runs, mitigating redundant mistakes. (iv) *Model Context Protocols*. These architectures liberally use MCPs to access data and tools, reducing the non-determinism in research systems. Frameworks like Paper2Agent (Miao et al. 2025) and DeepScientist (Weng et al. 2025) use MCP server interfaces to decouple the reasoning engine from domain-specific software tools (e.g., bioinformatics suites, GPU solvers, CLI utilities), turning static software and papers into standardized, callable agent tools.

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